

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I Prasanna / 190312642 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Prasanna

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

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Col - AT-2

1. Solution:-

Given network : 192.168.100.0/24

A/24 network contains 256 IP address (192.168.100.0 - 192.168.100.255).

Department :-

School of Computing, School of Electronics, School of Mechanical Engineering unused Range.

Prefix:-

/25, /26, /27, /28

Network address:-

192.168.100.0, 128, 192, 224

Host Range:-

192.168.100.1 - 192.168.100.126, 192.168.100.129 - 192.168.100.190,

192.168.100.193 - 192.168.100.222, 192.168.100.225 - 192.168.100.254.

Usable Hosts

126, 62, 30.

1. Network address and Broadcast address:-

School computing - 192.168.100.0 - 192.168.100.127

School of Electronics - 192.168.100.128 - 192.168.100.191

School of Mechanical - 192.168.100.192 - 192.168.100.223

Engineering

2. Valid Host IP address Range:-

School of computing - 192.168.100.1 - 192.168.100.126

3. number of usable host addresses :-

Formula :-

$$\text{usable hosts} = 2^x (\text{prefix length}) - 2$$

$$25 - 2^8 - 2 = 126 \text{ hosts}$$

$$26 - 2^8 - 2 = 62 \text{ hosts}$$

$$27 - 2^8 - 2 = 30 \text{ hosts}$$

4. Remaining unused IP address range :-

The allocated subnets company :- 192.168.100.0 - 192.168.100.123

The remaining unused addresses are :-

remaining range :- 192.168.100.0 - 192.168.100.255

These correspond to a /27 subnet.

Details :-

unused network address :- 192.168.100.254

valid host range :- 192.168.100.255 - 192.168.100.254

Broadcast address :- 192.168.100.255

usable hosts :- 30

2. A multinational software company :-

owner :-

Planned IP space :- 10.10.0.0/16

The /16 network includes

network address :- 10.10.0.0

Broadcast address :- 10.10.255.255

Total IP address :- 65536

ii) Network address and Broadcast address :-

Department :- software development, Quality Assurance, Human Resource

Creating management

Network address :-

10.10.0.0, 10.10.1.0, 10.10.1.128, 10.10.1.192

Broadcast address :-

10.10.0.255, 10.10.1.127, 10.10.1.191, 10.10.1.223

ii) valid host IP address range :-

Department :-

software development, Quality Assurance, Human Resource, Executive management.

valid host range :-

10.10.0.1 - 10.10.0.254, 10.10.1.1 - 10.10.1.126, 10.10.1.129 - 10.10.1.190

10.10.1.193 - 10.10.1.222

iii) no. of usable hosts :-

∴ usable hosts = $2^x (\text{prefix length}) - 2$

Department :-

software development, Quality Assurance, Human Resource, Executive management.

Prefix :- 124, 25, 26, 27

calculation :- $2^8 - 2$, $2^7 - 2$, $2^6 - 2$, $2^5 - 2$

usable hosts :- 254, 126, 62, 30

iv) Remaining unused IP address range :-

Allocated subnets use :-

10.10.0.0 - 10.10.1.223

The first unused address is :- 10.10.1.224

The remaining unused range is :

10.10.1.224 - 10.10.255.255

This includes :-

Remaining part of 10.10.1x :

10.10.1.224 - 10.10.1.255

Entire remaining address :-

10.10.2.0 - 10.10.255.255

3. A smart hospital :-

Given :-

IPv6 address :-

2001:db8:abcd:0000:0000:0000:0000:0000/64

i) Convert the IPv6 address into compressed notation :-

Rules for IPv6 compression :-

Remove leading zeros in each hexadecimal block

Replace one consecutive sequence of all zero blocks with ::

compressed IPv6 address :

2001:db8:abcd::/64

ii) Expand the compressed address back to full Hexadecimal Representation.

Expanding 2001:db8:abcd:: back to eight 16 bit blocks :

2001:db8:abcd:0000:0000:0000:0000:/64

iii) identify the network prefix using the /64 prefix length:-

A/64 prefix means:-

First 64 bits - network prefix

Last 64 bits - Interface identifier (Host ID)

Network prefix

2001:0db8:abcd:0::/64

iv) calculate the total number of Host addresses supported within the subnet

Since the prefix length is /64:

$$\text{Host bits} = 128 - 64 = 64 \text{ bits}$$

$$\text{Total host addresses:- } 2^{64} = 18446744073709551616$$

v) Tot. no. of possible Host addresses Available within this network

IPv6 does not reserve separate network and broadcast addresses as IPv4 does. Therefore, all 2^{64} addresses are usable.

4. An Internet service provider (ISP):-

Given:-

Routing table

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Subnet Mask

255.255.255.0

255.255.255.0

255

Destination IP address:- 192.168.2.45

ii) determine the destination subnet:-

0/24 subnet means the first 24 bits (first three octets)

destination IP address:- 192.168.2.45

network portion:- 192.168.2.0/24

Reason:- destination subnet: 192.168.2.0/24

iii) Matching Routing Table Entry:-

compare the destination network with the routing table

Routing Entry

Match

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

iv) Next-hop network / interface:-

Outgoing network: 192.168.2.0/24

Next Hop Interface: Interface connected to the 192.168.2.0/24

network (e.g., Gigabit Ethernet 0/1, depending).

v) Default Route:-

if the destination IP does not match any routing table entry,

the router uses the default route.

default route:- 0.0.0.0/0

The packet is forwarded to the default gateway (next-hop)

router) connected to the 192.168.10.126

5. A company:-

Example

advertisement LAN

network:- 192.168.10.0/26

gateway:- 192.168.10.1

Finance LAN

network:- 192.168.10.64/26

gateway:- 192.168.10.65

subnet mask for /26: 255.255.255.192

Each /26 subnet contains

64 IP addresses

62 usable host addresses.

ii) determine the network address and broadcast address.

LAN

Network address

Broadcast address

advertisement

192.168.10.0

192.168.10.63

Finance

192.168.10.64

192.168.10.127

iii) valid host IP address Range.

LAN

valid host Range

advertisement

192.168.10.1 - 192.168.10.62

Finance

192.168.10.65 - 192.168.10.126

i) Suggested IP addresses for three computers.

Computer	IP address
PC-1	192.168.10.2
PC-2	192.168.10.3
PC-3	192.168.10.4

Gateway 192.168.10.1

Finance LAN

Computer	IP address
PC-1	192.168.10.66
PC-2	192.168.10.67
PC-3	192.168.10.68

Gateway : 192.168.10.65

• Default Gateway for Each LAN :-

LAN	Default Gateway
Administration	192.168.10.1
Finance	192.168.10.65

The default gateway is the router interface that enables communication with other networks.